Sam McClenaghan

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Education

University of Victoria — Victoria, BC
Honours Bachelor of Science in Computer Science — *Sep 2023 – Dec 2027*
Minor in Mathematics | GPA 7.8/9

Experience

Software Engineer, Intern — May 2026 – Present
Vertical City — *Victoria, BC*

- Eliminated $20K in annual vendor costs by building an AWS media pipeline with S3 and CloudFront, serving 2M+ requests/day across 10,000+ screens
- Reduced media processing latency from 5 minutes to seconds by implementing event-driven, in-process rendering for near-real-time display updates
- Reduced bandwidth usage by 30% by developing a perceptual hashing system that identifies duplicate media across different encodings

Full-Stack Developer, Intern — May 2025 – Dec 2025
Leanpub — *Remote (Victoria, BC)*

- Improved accessibility coverage for 50,000+ ebooks by building an AI-powered image alt-text pipeline using AWS Lambda and Step Functions
- Shipped a production rich text editor for 50,000+ authors by integrating Lexical with React and TypeScript into core content workflows

Software Developer, Intern — Nov 2022 – Aug 2024
Trust Science — *Edmonton, AB*

- Reduced AWS S3 storage costs by 15% by implementing a file version-control system with HTTP/2 and JSON Patch for efficient delta storage
- Increased test coverage from 35% to 80% with Playwright E2E tests and Jest unit tests

Projects

CourseFlow | *TypeScript, React, Cloudflare Workers, D1* | courseflow.smccl.ca — 2025 – 2026

- Built and launched a course planner used by 2,000+ UVic students to create 3,000+ shareable timetables
- Developed a course recommendation engine combining embedding-based cosine similarity, BM25 lexical search, and course-level ranking, precomputing results in D1 to avoid request-time inference
- Implemented real-time schedule sharing using WebSockets and Cloudflare Durable Objects, broadcasting schedule changes and access revocations to connected viewers

Anchor | *Expo, React Native, TypeScript, Supabase, Anthropic SDK, Resend* — 2025

- Built a mental-health companion at the UVic Claude Hackathon, integrating Garmin HRV data and the Anthropic SDK to offer Claude-powered grounding exercises
- Deployed the React Native app to TestFlight and integrated Supabase backend services

Technical Skills

Languages: TypeScript, Python, JavaScript, Ruby, Go, Java, C, SQL
Development and Testing: React, Next.js, React Native, Ruby on Rails, Docker, Jest, Playwright
Cloud: AWS (S3, Lambda, MediaConvert, CloudFront, DynamoDB, EC2, Step Functions), BigQuery, Cloudflare Workers
Databases: MySQL, PostgreSQL, Redis, Elasticsearch, Supabase